

Normalization In-Class Activity

Lastname, Firstname

```
Registration(student_id, student_name, phone, address, major, course_id,
course_name, instructor, instructor_type, grade, location, time, session)
```

Candidate Keys

The only attribute combination that uniquely determines every other attribute in the relation is **(student_id, course_id)**. A student can register for many courses and a course has many students, so neither attribute alone is unique; grade specifically only makes sense for a given student in a given course. No other candidate key exists in the unnormalized relation.

Functional Dependencies

- `student_id → student_name, phone, address, major`
- `course_id → course_name, instructor, location, time, session`
- `instructor → instructor_type`
- `student_id, course_id → grade`

`student_id` and `course_id` are each **partial** determinants (they determine attributes using only part of the composite key). `instructor → instructor_type` is a **transitive** dependency, since `instructor_type` depends on `instructor`, which itself depends on `course_id` rather than on the key directly. Only `grade` is **fully** dependent on the whole key (`student_id, course_id`).

Convert to 2NF

2NF removes partial dependencies — attributes that depend on only part of the composite key. `student_name`, `phone`, `address`, and `major` depend only on `student_id`, and `course_name`, `instructor`, `location`, `time`, and `session` depend only on `course_id`, so each group is moved into its own relation.

Student	Course	Registration
<code>student_id</code> (PK)	<code>course_id</code> (PK)	<code>student_id</code> (PK, FK)
<code>student_name</code>	<code>course_name</code>	<code>course_id</code> (PK, FK)
<code>phone</code>	<code>instructor</code>	<code>grade</code>
<code>address</code>	<code>instructor_type</code>	
<code>major</code>	<code>location</code>	
	<code>time</code>	
	<code>session</code>	

This is still not 3NF: `instructor_type` depends on `instructor`, not directly on `course_id`, so `Course` still contains a transitive dependency.

Convert to 3NF

3NF removes transitive dependencies — non-key attributes that depend on another non-key attribute rather than directly on the primary key. Since `instructor_type` depends on `instructor` (not on `course_id`), `instructor` and `instructor_type` are moved into their own `Instructor` relation.

Student	Instructor	Course	Registration
student_id (PK)	instructor (PK)	course_id (PK)	student_id (PK, FK)
student_name	instructor_type	course_name	course_id (PK, FK)
phone		instructor (FK)	grade
address		location	
major		time	
		session	

Final 3NF relations:

- `Student(student_id, student_name, phone, address, major)`
- `Instructor(instructor, instructor_type)`
- `Course(course_id, course_name, instructor, location, time, session), FK instructor → Instructor`
- `Registration(student_id, course_id, grade), FK student_id → Student, FK course_id → Course`

Crow's Foot ERD

Registration System — Crow's Foot ERD (3NF)

